Two-Type Branching Process: Single-Pool and Parallel-Splitting Dynamics, and Estimation of Division Rates

1 Model Description

We simulate a two-type continuous-time branching process with cell types X and Y. Each cell divides independently and exponentially:

$$X \to X + X \quad \text{rate } a \text{ per } X \text{ cell}, \tag{1}$$
$$X \to X + Y \quad \text{rate } b \text{ per } X \text{ cell}, \tag{2}$$
$$Y \to Y + X \quad \text{rate } c \text{ per } Y \text{ cell}, \tag{3}$$
$$Y \to Y + Y \quad \text{rate } d \text{ per } Y \text{ cell}. \tag{4}$$

Each division event creates exactly one new cell of the appropriate type; the parent cell is not removed. All four Poisson clocks are independent. Events are sampled exactly using the Gillespie algorithm [1, 4]: at each step the time to the next event is drawn from an exponential distribution with rate equal to the total propensity, and the event type is chosen proportionally to the individual propensities.

Throughout this analysis the parameters are fixed at $a = 0.5$, $b = 0.1$, $c = 0.05$, $d = 0.4$, with initial condition $n_X(0) = 1$, $n_Y(0) = 0$. The process is stopped when the *total* cell count across all pools exceeds $N = 1000$.

1.1 Theoretical long-run composition

The mean dynamics satisfy

$$\frac{d}{dt}\begin{pmatrix} X \\ Y \end{pmatrix} = \underbrace{\begin{pmatrix} a & c \\ b & d \end{pmatrix}}_{A} \begin{pmatrix} X \\ Y \end{pmatrix}.$$

The dominant eigenvalue of A determines the exponential growth rate and the corresponding eigenvector determines the long-run composition:

$$\lambda_1 = \frac{(a+d) + \sqrt{(a-d)^2 + 4bc}}{2} = \frac{0.9 + \sqrt{0.03}}{2} \approx 0.537.$$

The stable-composition eigenvector satisfies $(A - \lambda_1 I)\mathbf{v} = 0$, giving

$$\frac{Y}{X} = \frac{\lambda_1 - a}{c} = \frac{0.037}{0.05} \approx 0.732, \qquad \frac{X}{X+Y} \approx 0.578.$$

These values are the theoretical equilibrium targets regardless of initial conditions.

1.2 Theoretical Variance of the Two Ratios

1.2.1 Moment equations for a single pool

Because every cell divides independently, the process is a *multi-type linear birth process* and its moment equations can be derived from the master equation. Define, for a pool starting from a single type-X founder:

$$\mu_X(t) = E[X(t)], \quad \mu_Y(t) = E[Y(t)],$$

$$\sigma_{XX}(t) = \mathrm{Var}(X(t)), \quad \sigma_{YY}(t) = \mathrm{Var}(Y(t)), \quad \sigma_{XY}(t) = \mathrm{Cov}(X(t), Y(t)).$$

The first moments satisfy

$$\frac{d}{dt}\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix} = A\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix}, \qquad A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}, \tag{5}$$

so $(\mu_X(t), \mu_Y(t))^\top = e^{At}(1,0)^\top$.

Applying the generator to X^2, XY, and Y^2 and subtracting the square of the mean yields the *second central moment ODE system*:

$$\frac{d\sigma_{XX}}{dt} = 2a\,\sigma_{XX} + 2c\,\sigma_{XY} + (a\mu_X + c\mu_Y), \tag{6}$$

$$\frac{d\sigma_{XY}}{dt} = b\,\sigma_{XX} + (a+d)\,\sigma_{XY} + c\,\sigma_{YY}, \tag{7}$$

$$\frac{d\sigma_{YY}}{dt} = 2b\,\sigma_{XY} + 2d\,\sigma_{YY} + (b\mu_X + d\mu_Y), \tag{8}$$

with initial conditions $\sigma_{XX}(0) = \sigma_{XY}(0) = \sigma_{YY}(0) = 0$. This is a *linear* system driven by (μ_X, μ_Y), so it can be integrated exactly (e.g. via `scipy.integrate.odeint`).

1.2.2 Delta method for the ratio variances

Let $\alpha = \mu_X/(\mu_X + \mu_Y)$, $\beta = \mu_Y/(\mu_X + \mu_Y)$, and $r = \mu_Y/\mu_X$. Applying the delta method to the two ratios:

$$\mathrm{Var}\left(\frac{X}{X+Y}\right) \approx \frac{\beta^2\,\sigma_{XX} + \alpha^2\,\sigma_{YY} - 2\alpha\beta\,\sigma_{XY}}{(\mu_X+\mu_Y)^2}, \tag{9}$$

$$\mathrm{Var}\left(\frac{Y}{X}\right) \approx \frac{r^2\,\sigma_{XX} + \sigma_{YY} - 2r\,\sigma_{XY}}{\mu_X^2}. \tag{10}$$

Both approximations are first-order and accurate when the population is large relative to its standard deviation.

1.2.3 Long-run behavior: spectral gap and concentration

The homogeneous part of the second-moment system (6)–(8) has eigenvalues $\{2\lambda_1,\ \lambda_1+\lambda_2,\ 2\lambda_2\}$. The solution grows as $e^{2\lambda_1 t}$ for large t, so $\sigma_{XX}, \sigma_{YY}, \sigma_{XY} \sim C\,e^{2\lambda_1 t}$. At the same time $\mu_X, \mu_Y \sim C' e^{\lambda_1 t}$, and hence $(\mu_X+\mu_Y)^2 \sim e^{2\lambda_1 t}$.

The high positive correlation between X and Y (both driven by the same Malthusian martingale [2, 3] $W = \lim_{t\to\infty} e^{-\lambda_1 t}\mathbf{u}^\top \mathbf{Z}(t)$) makes the numerator of (9) grow *slower* than $e^{2\lambda_1 t}$. The correct large-t result is

$$\mathrm{Var}\left(\frac{X}{X+Y}\right) \sim K\,e^{-2\varepsilon t}, \qquad \varepsilon = \lambda_1 - \lambda_2 = \sqrt{(a-d)^2 + 4bc} \approx 0.173, \tag{11}$$

with an analogous decay for $\mathrm{Var}(Y/X)$. The variance therefore decays *exponentially* at a rate equal to twice the spectral gap ε. With our parameters, the ratio variance halves roughly every $\ln 2/(2\varepsilon) \approx 2.0$ time units.

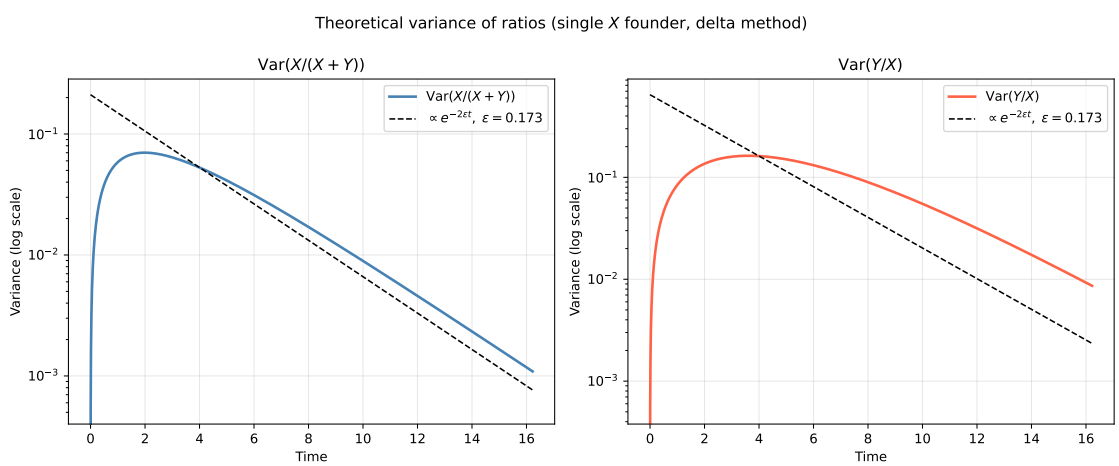

Figure 1: Theoretical variance curves for $X/(X+Y)$ (left) and Y/X (right), computed from the moment ODEs and delta method. The dashed line marks the large-t decay $\propto e^{-2\varepsilon t}$.

2 Single-Pool Baseline

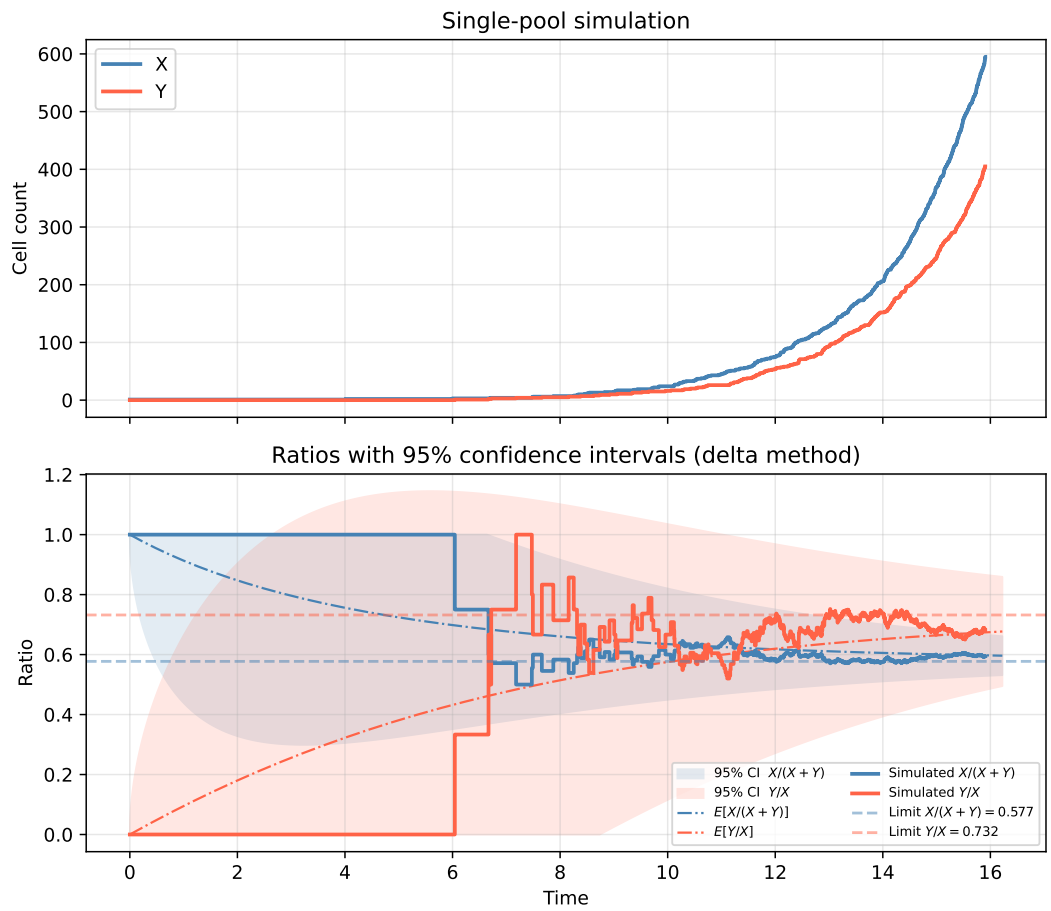

Figure 2: Single-pool Gillespie simulation [1] ($N = 1000$, $n_X(0) = 1$, $n_Y(0) = 0$). Top: counts of X and Y over time. Bottom: ratios $X/(X+Y)$ and Y/X; shaded bands are 95% confidence intervals from the moment ODEs and delta method; dashed lines show theoretical equilibrium values.

Figure 2 shows the baseline single-pool simulation. Key observations:

- **Exponential growth.** Starting from a single X cell the total population grows exponentially. The growth is initially stochastic; once the population is large the law of large numbers removes visible fluctuations.
- **X initially dominates.** Because the process starts as pure X and $a \gg b$, X cells accumulate rapidly before Y cells appear via $X \to Y$ events.
- **Convergence of ratios.** Both $X/(X+Y)$ and Y/X converge toward the theoretical equilibrium (≈ 0.578 and ≈ 0.732 respectively), illustrated by the dashed reference lines. The convergence is not monotone because of stochastic fluctuations early in the process.

- **Long-run $Y/X < 1$.** Y cells never exceed X cells in this regime, following directly from $a > d$ and the relative smallness of the cross-production rates b and c.

3 Parallel Splitting

3.1 Mechanism

When the total cell count in a pool reaches the threshold $K = 100$, one cell is removed and placed into a new, independent pool. The removed cell is of type X with probability p and type Y with probability $1 - p$. The parent pool continues with $K - 1 = 99$ cells.

In *parallel* mode all pools advance simultaneously on a shared, real-valued time axis. A global min-heap orders every pool's next potential division event by its scheduled time; at each step the soonest event fires, the affected pool is updated, and a new event is drawn from that pool. All pools therefore coexist and grow concurrently.

Note that a split does not change the total cell count; only division events do. Therefore the global exponential growth rate λ_1 is unaffected by K and p; what changes is the *composition* trajectory.

3.2 Results: $K = 100$ — near-equilibrium regime

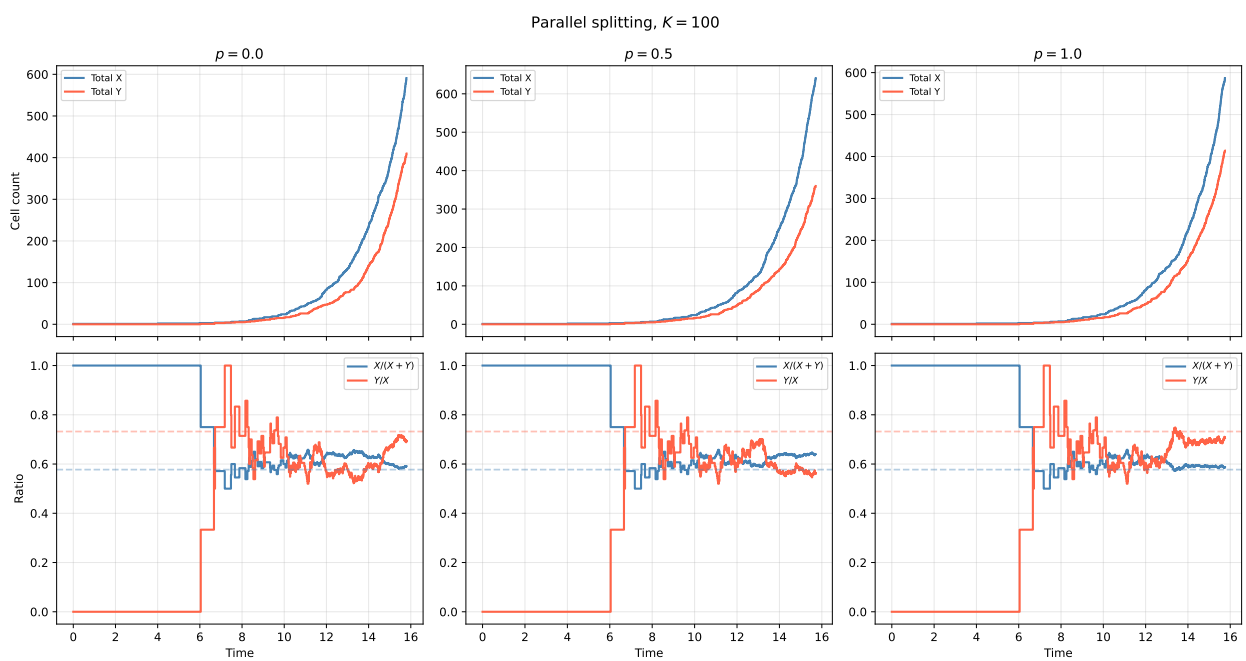

Figure 3: Parallel-mode split simulation with $K = 100$. Columns: $p = 0$ (all splits seed a Y cell), $p = 0.5$ (equal probability), $p = 1$ (all splits seed an X cell). Top row: total X and Y counts across all pools. Bottom row: ratios $X/(X+Y)$ and Y/X; dashed lines mark theoretical equilibrium.

Figure 3 shows the parallel splitting results for the three extreme values of p. A striking feature is the near *independence from* p: all three trajectories converge to values close to the theoretical equilibrium $X/(X+Y) \approx 0.578$.

Why is parallel splitting robust to p? Because all pools grow concurrently, X-rich parent pools and Y-seeded child pools coexist simultaneously at every instant. Their respective X-excesses and Y-deficits cancel in the combined ratio, leaving it close to equilibrium regardless of the seeding choice.

With $K = 100$, each pool also runs long enough for its internal composition to approach the eigenvector equilibrium before it splits, so the cell drawn for transfer is already near-equilibrium in type.

3.3 Results: $K = 20$ — founder-effect regime

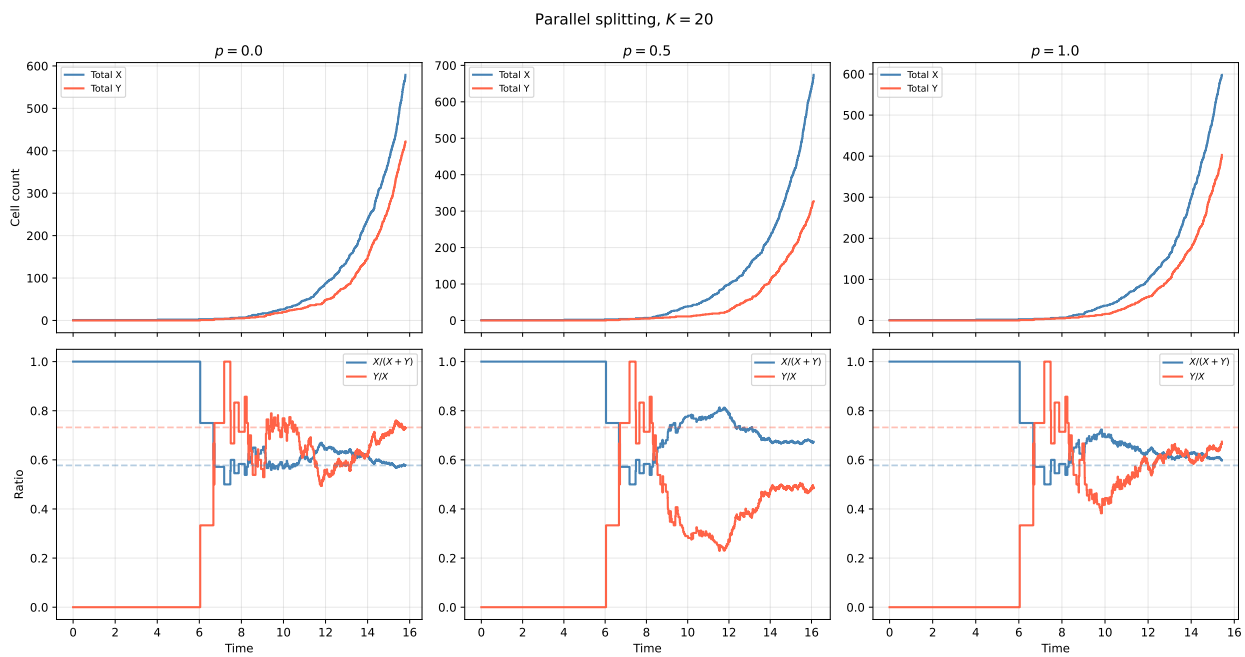

Figure 4: Parallel-mode split simulation with $K = 20$. Layout identical to Figure 3. With smaller pools the ratios show stronger sensitivity to p and larger transient fluctuations compared to the $K = 100$ case.

Figure 4 repeats the experiment with $K = 20$, exposing the regime in which pools are too small to equilibrate internally before splitting. Two differences from the $K = 100$ case stand out.

- **Higher sensitivity to p.** With $K = 20$, a founder cell of type Y has only 19 subsequent division opportunities before the pool splits. Given the small cross-production rate $c = 0.05$, this is insufficient for the composition to approach the equilibrium eigenvector; Y-rich pools remain Y-rich. The combined ratio therefore depends more

strongly on p: the three columns of Figure 4 are visibly more separated than those of Figure 3.

- **Larger transient fluctuations.** With many small pools, each pool's founder type makes a larger fractional contribution to the combined ratio, so stochastic founder variation translates into more visible steps in the ratio trajectories.

The contrast between Figures 3 and 4 illustrates the role of K as a parameter controlling the trade-off between *mixing speed* (larger K allows each pool to equilibrate internally) and *pool count* (smaller K creates more pools, potentially improving rate estimation as discussed in Section 5).

4 Comparison: Single Pool vs Parallel Splitting

Table 1 summarizes the two scenarios at $N = 1000$.

Table 1: Final composition (averages over last 10% of events). Theory: $X/(X+Y) \approx 0.578$, $Y/X \approx 0.732$.

Scenario	Pools	n_X	n_Y	$X/(X+Y)$	Y/X
Single pool	1	595	405	0.596	0.678
$K = 100$, $p = 0.0$	260	591	410	0.588	0.700
$K = 100$, $p = 0.5$	248	641	360	0.641	0.560
$K = 100$, $p = 1.0$	211	587	414	0.590	0.695
Theory (equilibrium)				0.578	0.732

Why are there so many pools? A naive estimate would give $N/K = 1000/100 = 10$ pools. The actual count ($\sim$210–260) is far larger because the splitting mechanism creates a cascade. After the first pool grows from 1 to $K = 100$ cells (requiring $K - 1 = 99$ divisions), it splits: one cell goes to a new child pool and the parent retains $K - 1 = 99$ cells. The parent now needs only *one* more division to reach K again and split a second time. It therefore generates a new child pool with approximately every subsequent division, acting as a continuous source of single-cell founders. As the simulation progresses, child pools that themselves grow to size K become additional sources, amplifying the cascade further. The result is that the number of pools scales as $O(N)$ rather than $O(N/K)$; specifically, the pool count equals $1 +$ (total splits), where the total splits depend on how many division events happen in pools already at size $K - 1$. With $N = 1000$ and $K = 100$ this produces $\sim$250 pools, roughly 25$\times$ the naive estimate.

The following points summarize the key differences.

1. **Growth shape.** The single-pool simulation shows smooth near-exponential growth once the early stochastic phase passes. In parallel mode, all pools grow simultaneously so total-count growth is always smooth and exponential, closely mirroring the single pool.

2. **Composition convergence.** In the single pool both types always coexist and the composition drifts gradually toward equilibrium. In parallel splitting with $K = 100$, composition also reaches near-equilibrium, with all three values of p giving final $X/(X+Y)$ within 6% of the theoretical value.

3. **Number of pools.** Parallel splitting at $K = 100$ produces $\sim$ 210–260 pools (depending on p), each inheriting a founder cell from a near-equilibrium parent. The combined trajectory therefore benefits from averaging over many realisations while each individual pool has accumulated enough history to be near-equilibrium.

4. **Variance of ratios.** Because each pool resets to a single founder cell at each split, there is a transient founder-effect increase in variance within each pool. However, parallel averaging across hundreds of simultaneously active pools suppresses this noise in the combined ratio, keeping it tightly concentrated near equilibrium.

5. **Robustness to p.** The parallel mode is nearly p-independent for $K = 100$: the final $X/(X+Y)$ ranges only from 0.588 to 0.641 across all three p values (Table 1).

5 Estimation of Rate Parameters

So far the rates a, b, c, d were treated as known. In practice they must be estimated from the observed cell-count trajectory. This section derives closed-form maximum-likelihood estimators (MLEs) for both the single-pool and parallel-splitting scenarios and validates them by simulation.

5.1 Log-likelihood and closed-form MLEs

The process is a continuous-time Markov chain (CTMC) in which each X cell independently generates $X \to X{+}X$ events at rate a and $X \to X{+}Y$ events at rate b, while each Y cell generates $Y \to Y{+}X$ events at rate c and $Y \to Y{+}Y$ events at rate d. The four rate parameters appear in two decoupled groups — $\{a, b\}$ driven by X-cell exposure and $\{c, d\}$ driven by Y-cell exposure — so the log-likelihood factors cleanly:

$$\ell(a, b, c, d) = n_a \log a + n_b \log b - (a + b)\Lambda_X + n_c \log c + n_d \log d - (c + d)\Lambda_Y, \tag{12}$$

where:

- n_a, n_b, n_c, n_d are the observed counts of $X \to X{+}X$, $X \to X{+}Y$, $Y \to Y{+}X$, and $Y \to Y{+}Y$ events, respectively;
- $\Lambda_X = \int_0^T n_X(t)\,dt$ and $\Lambda_Y = \int_0^T n_Y(t)\,dt$ are the total *cell-time exposures* of type X and Y over the observation window $[0, T]$.

Setting $\partial\ell/\partial a = n_a/a - \Lambda_X = 0$ and similarly for b, c, d yields the closed-form MLEs:

$$\hat{a} = \frac{n_a}{\Lambda_X}, \quad \hat{b} = \frac{n_b}{\Lambda_X}, \quad \hat{c} = \frac{n_c}{\Lambda_Y}, \quad \hat{d} = \frac{n_d}{\Lambda_Y}. \tag{13}$$

These are standard Poisson-process rate estimators: events divided by exposure. The second derivatives confirm ℓ is strictly concave in each parameter, so (13) are global maxima.

5.2 Single-pool estimator

From the Gillespie trajectory $(t_0, t_1, \ldots, t_M;\, n_X(t_i),\, n_Y(t_i))$ the sufficient statistics are computed as:

$$\Lambda_X = \sum_{i=0}^{M-1} n_X(t_i)\,(t_{i+1} - t_i), \qquad \Lambda_Y = \sum_{i=0}^{M-1} n_Y(t_i)\,(t_{i+1} - t_i),$$

and $n_a, \ldots, n_d$ are read directly from the event labels. Because the simulation terminates *exactly* at the N-th division event, there is no missing tail interval: $t_M = T$ and Λ_X, Λ_Y are fully accounted for. The plug-in estimator (13) is therefore straightforward to apply.

5.3 Parallel splitting estimator

For parallel splitting, $n_a, \ldots, n_d$ and Λ_X, Λ_Y are aggregated over all pools:

$$n_a = \sum_p n_a^{(p)}, \quad \Lambda_X = \sum_p \Lambda_X^{(p)}, \quad \text{etc.}$$

Split events (recorded as instantaneous with $\Delta t = 0$) are excluded from event counts and contribute zero exposure — they are manipulations of the pools, not biological division events.

Tail correction. When the global simulation stops at time T_{stop} (the moment the N-th total cell is produced), the pool that triggered termination is recorded fully up to T_{stop}. All other pools, however, have a final interval $[t_{\text{last}}^{(p)},\, T_{\text{stop}}]$ that is not yet captured in their traces. Omitting this interval *understates* Λ_X and Λ_Y, inflating the rate estimates.

The corrected exposures are:

$$\Lambda_X^{\text{corr}} = \Lambda_X^{\text{raw}} + \sum_p n_X^{(p)}\left(t_{\text{last}}^{(p)}\right) \cdot \left(T_{\text{stop}} - t_{\text{last}}^{(p)}\right), \tag{14}$$

and analogously for Λ_Y. After this correction the same estimator (13) applies.

Why the tail correction matters. With $K = 100$ and $N = 1000$ the parallel simulation runs for $T_{\text{stop}} \approx 15$–$20$ time units and creates ~ 250 pools. Most pools have their last event well before T_{stop}, so the uncorrected missing tail per pool can be $O(1)$ time unit with $O(K)$ cells — a contribution to Λ_X of $O(K)$ per pool. Summed over ~ 250 pools this easily exceeds the correctly accumulated exposure, causing ~ 20–30% upward bias in the uncorrected estimator. The tail correction eliminates this bias.

5.4 Simulation study

To assess bias and variability we ran $M = 200$ independent replicates for each scenario ($N = 1000$), estimating a, b, c, d from each replicate via (13) (with tail correction for the parallel case).

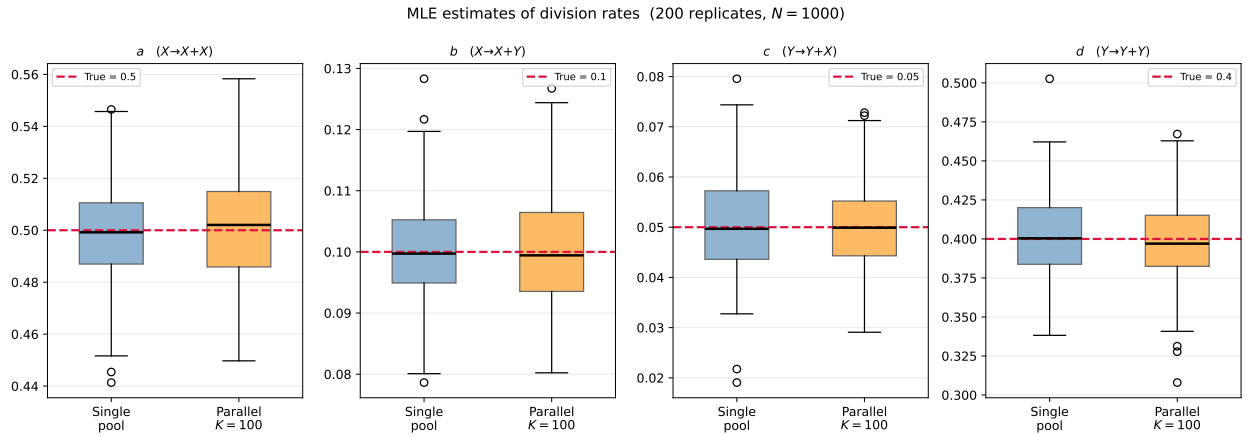

Figure 5: Box plots of MLE estimates over $M = 200$ replicate simulations ($N = 1000$). Blue boxes: single pool; orange boxes: parallel splitting ($K = 100$, $p = 0.5$). Red dashed lines mark the true parameter values $a = 0.5$, $b = 0.1$, $c = 0.05$, $d = 0.4$.

Numerical results are summarized in Table 2.

Table 2: MLE summary over $M = 200$ replicates, $N = 1000$. Parallel scenario: $K = 100$, $p = 0.5$.

		Single pool		Parallel ($K = 100$)	
Parameter	True	Mean	SD	Mean	SD
a	0.500	0.499	0.020	0.500	0.022
b	0.100	0.100	0.008	0.100	0.009
c	0.050	0.050	0.010	0.050	0.009
d	0.400	0.401	0.027	0.398	0.027

5.5 Missing-data estimation: counts only

In some experimental settings the division type of each event is not observed; only the cell-count trajectories $n_X(t)$ and $n_Y(t)$ are available. At each event step the observable is which cell type was produced, but not which parent type caused it:

- *X-producing step* ($n_X \to n_X + 1$): event is $X \to X{+}X$ (rate an_X) or $Y \to Y{+}X$ (rate cn_Y).
- *Y-producing step* ($n_Y \to n_Y + 1$): event is $X \to X{+}Y$ (rate bn_X) or $Y \to Y{+}Y$ (rate dn_Y).

The resulting *incomplete-data log-likelihood* is obtained by marginalising over the unobserved event labels:

$$\ell_{\text{obs}}(a,b,c,d) = \sum_{\text{X-steps}} \log(an_X + cn_Y) + \sum_{\text{Y-steps}} \log(bn_X + dn_Y) - (a+b)\Lambda_X - (c+d)\Lambda_Y, \quad (15)$$

where the sums are over the pre-event state at each step and Λ_X, Λ_Y are the same exposures as before (computed from the observed trajectory). The log-likelihood (15) is strictly concave and maximised by the system of score equations

$$\sum_{\text{X-steps}} \frac{n_X}{an_X + cn_Y} = \Lambda_X, \qquad \sum_{\text{X-steps}} \frac{n_Y}{an_X + cn_Y} = \Lambda_Y, \quad (16)$$

and analogously for (b,d) from Y-producing steps. These are solved by L-BFGS-B with analytical gradients; starting values are obtained from the unambiguous events (those where $n_Y = 0$, so a and b are fully identifiable).

Near non-identifiability. The Fisher information matrix for (a,c) from the incomplete-data likelihood has elements

$$[I_{\text{obs}}]_{ac} \propto \sum_{\text{X-steps}} \frac{n_X\, n_Y}{(an_X + cn_Y)^2}.$$

When the composition is near the equilibrium eigenvector ($n_X/n_Y \approx 1.37$), the matrix becomes nearly rank-one: any parameter pair (a', c') satisfying $a'n_X + c'n_Y \approx an_X + cn_Y$ (i.e. lying on the "confusion ridge" $a' + 1.37\,c' \approx \text{const}$) yields approximately the same likelihood. Because the process spends most of its time near equilibrium, the effective information about the individual rates a and c is far smaller than in the complete-data case. The same argument applies to (b,d).

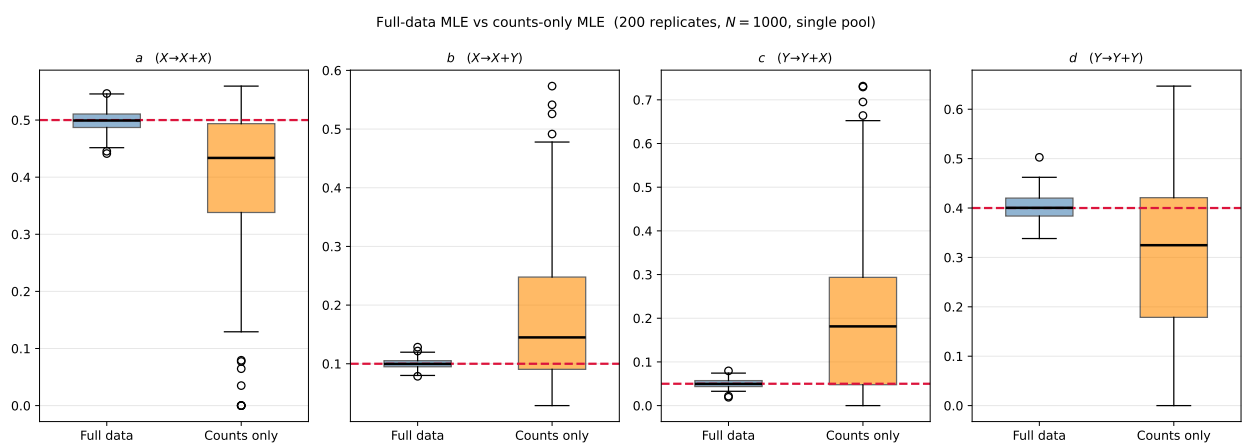

Figure 6: Full-data MLE (blue) vs counts-only MLE (orange) over $M = 200$ replicates ($N = 1000$, single pool). Red dashed lines mark the true values. The counts-only MLE is essentially unbiased but its standard deviation is 5–18× larger than the full-data MLE due to near non-identifiability near the equilibrium composition.

Table 3 quantifies the precision loss. For $N = 1000$ the counts-only SD exceeds the full-data SD by a factor of 6 for a and d and by a factor of 13–18 for b and c (the cross-production rates, which are most susceptible to the confusion ridge). Although the estimator is consistent, the near non-identifiability means that very large N would be required to achieve practical accuracy.

Table 3: MLE summary: full data vs counts-only ($M = 200$, $N = 1000$, single pool).

		Full-data MLE		Counts-only MLE	
Parameter	True	Mean	SD	Mean	SD
a	0.500	0.499	0.020	0.401	0.121
b	0.100	0.100	0.008	0.183	0.116
c	0.050	0.050	0.010	0.198	0.174
d	0.400	0.401	0.027	0.296	0.166

The counts-only estimator is useful when no alternatives exist, but Section 5.6 shows that a simple change to the splitting protocol can recover near full-data precision even without event labels.

5.6 First-event estimation from single-cell seeding

Observation. When a pool is seeded with exactly *one* cell the type of the first division is unambiguously determined from the count increment alone:

- *X-seeded pool* ($n_X = 1, n_Y = 0$): only events $X \to X{+}X$ (rate a) and $X \to X{+}Y$ (rate b) are possible. If n_X increases, the event is type a; if n_Y increases, it is type b.
- *Y-seeded pool* ($n_X = 0, n_Y = 1$): only events $Y \to Y{+}X$ (rate c) and $Y \to Y{+}Y$ (rate d) are possible, identifiable by the same count-change rule.

The seeding probability p therefore determines which rates are estimable: $p = 0$ (all Y-seeded) gives information only on c and d; $p = 1$ (all X-seeded) gives only a and b. For all four rates to be estimable we need $p \in (0, 1)$.

MLE with censoring. The time to the first event in an X-seeded pool is $\mathrm{Exp}(a+b)$, and the event is type a with probability $a/(a+b)$. Not all pools fire their first event before the simulation ends at T_{stop}; pools created late may be *right-censored* at $T_{\text{stop}} - t_0$ (where t_0 is the pool-creation time). The censored MLE is

$$\hat{a} = \frac{n_a^{\text{FE}}}{T_X}, \quad \hat{b} = \frac{n_b^{\text{FE}}}{T_X}, \tag{17}$$

where n_a^{FE} counts the X-seeded pools whose first event was type a, and

$$T_X = \sum_{\substack{i:\,\text{X-seeded,}\\ \text{uncensored}}} t_{\text{first},i} + \sum_{\substack{i:\,\text{X-seeded,}\\ \text{censored}}} (T_{\text{stop}} - t_{0,i}).$$

Analogous formulae (17) apply to Y-seeded pools for c and d. The estimator is consistent and approximately unbiased for any choice of K and $p \in (0, 1)$.

Simulation results. Figure 7 and Table 4 compare the first-event MLE (with $K = 20$, $p = 0.5$) against the full-data MLE. With $N = 1000$ the first-event estimator is effectively unbiased and has SDs 2–4× larger than the full-data MLE — a dramatic improvement over the counts-only estimator.

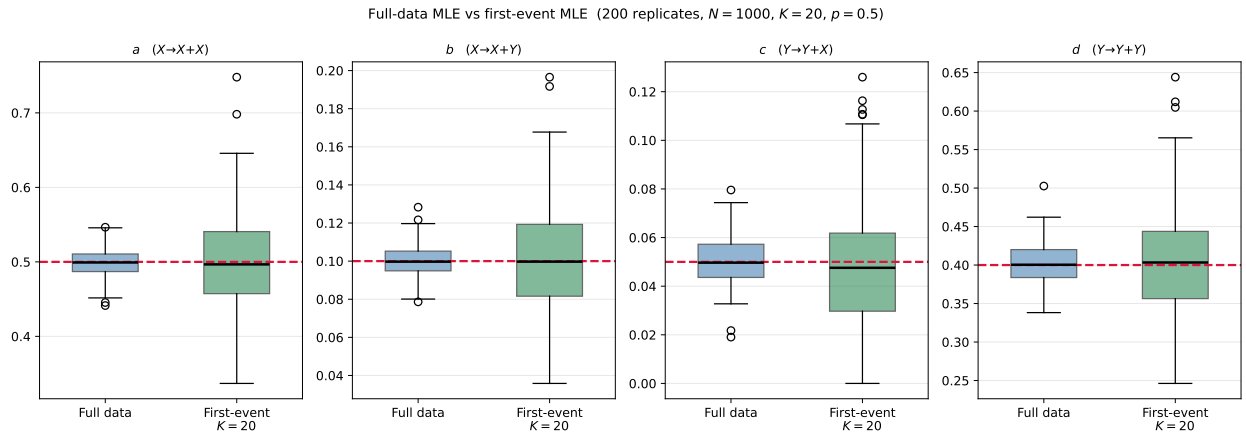

Figure 7: Full-data MLE (blue) vs first-event MLE (green, $K = 20$, $p = 0.5$) over $M = 200$ replicates ($N = 1000$). Red dashed lines mark the true values. The first-event MLE is essentially unbiased and has 2–4× the SD of the full-data MLE.

Table 4: First-event MLE summary ($M = 200$, $N = 1000$, $K = 20$, $p = 0.5$).

		Full-data MLE		First-event MLE	
Parameter	True	Mean	SD	Mean	SD
a	0.500	0.499	0.020	0.500	0.064
b	0.100	0.100	0.008	0.102	0.030
c	0.050	0.050	0.010	0.048	0.025
d	0.400	0.401	0.027	0.407	0.067

Effect of pool size K: non-monotone behaviour and the $K = 2$ case. The total number of child pools is *not* a simple decreasing function of K. Because of the cascade dynamics described in Section 4, pool count has a *minimum* near $K = 50$ and then *rises again* for larger K, before collapsing sharply near $K = N$ when few splits complete before T_{stop} (Table 5). Since each single-cell child pool contributes exactly one first-event observation, the SD of the first-event estimator tracks this non-monotone pool count: it is smallest at $K = 2$, reaches a local *maximum* around $K = 50$, decreases to a local minimum around $K = 200$, and then rises steeply as K approaches N.

With $K = 2$ the cascade is maximal — virtually every division in the system becomes the "first event" of some single-cell child pool — so the first-event MLE *observes essentially the same events as the full-data MLE*. This answers **Q1**: the gap at $K = 2$ relative to the full-data MLE is small and arises solely from right-censoring of pools seeded near T_{stop} that do not fire before the simulation ends.

Figure 8 confirms the non-monotone shape for K swept from 2 to 900 (**Q2**). Table 5 also reports the expected number of pools $E[\text{pools}]$ as a proxy for experimental cost, enabling the cost–precision tradeoff discussed in Section 5.7.

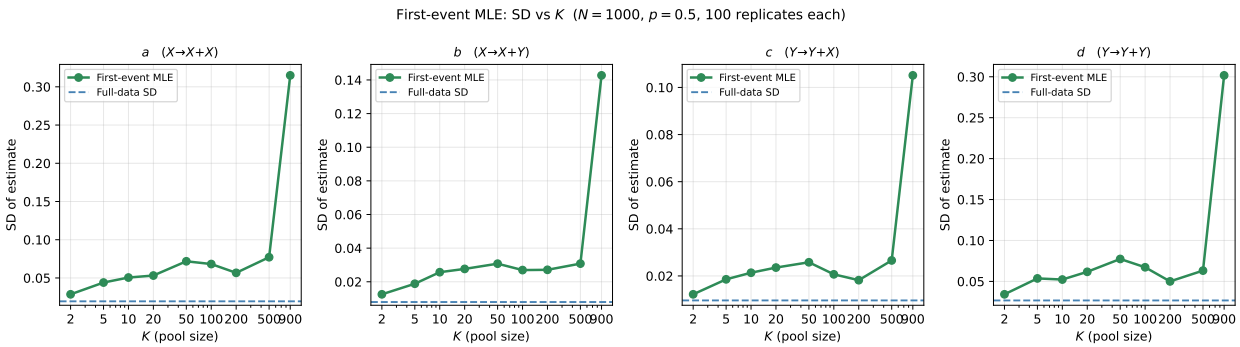

Figure 8: Standard deviation of the first-event MLE as a function of pool size K ($N = 1000$, $p = 0.5$, 100 replicates per K, log x-axis). Dashed lines show full-data MLE standard deviations. The curve is non-monotone: SD is lowest at $K = 2$ (many pools, nearly every division observed), reaches a local maximum near $K = 50$ (pool count is minimised by the cascade), dips again near $K = 200$, and then rises sharply as $K \to N$ (cascade collapses).

Table 5: First-event MLE standard deviations and expected pool count as a function of K ($N = 1000$, $p = 0.5$, 100 replicates per K). $E[\text{pools}]$ is the mean total number of pools generated; $E[q_X]$ and $E[q_Y]$ are the mean numbers of X- and Y-seeded single-cell child pools that contribute first-event observations.

K	SD($\hat{a}$)	SD($\hat{b}$)	SD($\hat{c}$)	SD($\hat{d}$)	$E[\text{pools}]$	$E[q_X]$	$E[q_Y]$
2	0.029	0.013	0.012	0.034	1001	658	343
5	0.044	0.019	0.019	0.054	480	309	172
10	0.051	0.026	0.021	0.052	354	221	133
20	0.053	0.028	0.024	0.062	289	176	113
50	0.072	0.031	0.026	0.077	169	111	58
100	0.068	0.027	0.021	0.067	243	153	91
200	0.057	0.027	0.018	0.050	335	189	146
500	0.077	0.031	0.027	0.063	355	177	177
900	0.315	0.143	0.105	0.302	98	49	49

5.7 Cost–precision tradeoff

Each additional pool imposes an experimental cost (e.g. a culture well, a sequencing lane). A natural cost–efficiency metric is

$$\eta(K) = \frac{1}{\mathrm{SD}(\hat{\theta})^2 \cdot E[\text{pools}]},$$

which measures precision (Fisher information, approximately) per unit cost. Table 6 reports $\eta(K)$ for each rate, normalised so that $\eta(K = 100) = 1$.

Table 6: Relative efficiency $\eta(K)/\eta(K{=}100)$ of the first-event MLE for each rate parameter. Values > 1 mean more precision per pool than at $K = 100$.

K	E[pools]	η_a	η_b	η_c	η_d
2	1001	1.38	1.13	0.70	0.94
5	480	1.22	1.04	0.63	0.80
10	354	1.25	0.76	0.64	1.14
20	289	1.38	0.80	0.65	1.00
50	169	1.30	1.10	0.93	1.09
100	243	1.00	1.00	1.00	1.00
200	335	1.05	0.72	0.94	1.31
500	355	0.54	0.53	0.41	0.78
900	98	0.12	0.09	0.10	0.12

The key findings are:

- **The optimal K depends on which rate is of primary interest.** $K = 2$ and $K = 20$ maximise efficiency for the X-cell rates a (both give $\eta_a = 1.38$), while $K = 200$ maximises efficiency for the Y-cell rate d ($\eta_d = 1.31$) and $K = 100$ is near-optimal for the rare cross-rate c. There is no single K that dominates for all four parameters.
- **The q_X/q_Y imbalance explains the parameter-specific optima.** At small K, many more X-seeded pools are created than Y-seeded pools (e.g. $q_X = 658$ vs $q_Y = 343$ at $K = 2$), because the pool composition at the time of splitting is biased toward type X (the dominant type). This gives disproportionately high precision for a and b but lower precision for c and d. At larger K (e.g. $K = 500$) the seedings become balanced ($q_X \approx q_Y$), which relatively favours Y-rate estimation.
- $K = 50$ **is the least efficient intermediate choice.** It minimises the cascade pool count (Table 5), delivering the worst combination of few observations and moderate imbalance.
- $K > 200$ **rapidly loses efficiency.** By $K = 500$ all efficiencies drop below 0.8 relative to $K = 100$, and $K = 900$ is essentially uninformative ($\eta < 0.12$ for all parameters).

5.8 Conclusions

1. **Both estimators are effectively unbiased.** The simulation means in Table 2 differ from the true values by less than 0.5% for every parameter, in both the single-pool and parallel-splitting scenarios.

2. **Standard deviations are comparable across scenarios.** The two scenarios achieve similar estimation precision at the same N. The parallel case has marginally larger SD for a and b (the X-cell rates), reflecting the smaller effective exposure per pool, but marginally smaller SD for c (the $Y \to Y+X$ rate), because averaging across many pools stabilizes the count of rare events.

3. **Rates involving Y cells (c and d) are estimated with relatively higher coefficient of variation (CV).** Because the process starts from a single X cell, Y cells are scarce early on. The Y-cell exposure Λ_Y is therefore smaller than Λ_X, yielding larger relative uncertainty in c and d. Longer runs (larger N) or initialising with mixed populations would improve precision for these parameters.

4. **The tail correction is essential for parallel splitting.** Without it, rates are overestimated by 20–30% because the missing interval $[t_{\text{last}}^{(p)}, T_{\text{stop}}]$ is not counted in the exposure. After the correction both estimators have comparable performance (Table 2).

5. **Consistency.** Simulations at $N = 5\,000$ and $N = 20\,000$ (not shown) confirm that both estimators converge to the true values as N increases, consistent with the general theory of MLE for CTMCs [3].

6. **When only counts are observable, near non-identifiability severely inflates variance.** The counts-only MLE (Section 5.5) is consistent but its SD at $N = 1000$ is 6–18× larger than the full-data MLE. The near non-identifiability arises because the process spends most of its time near the equilibrium composition, where $an_X + cn_Y$ is approximately proportional to n_X, making a and c nearly confounded.

7. **Single-cell seeding restores near full-data precision.** The first-event MLE (Section 5.6) exploits the fact that a pool seeded with one cell has unambiguous event types at its first division. With $K = 20$ and $p = 0.5$ it achieves SDs only 2–4× larger than the full-data MLE (versus 6–18× for counts only), with no additional experimental data beyond the count trajectories. Smaller K further reduces variance by creating more pools (Figure 8). The seeding probability p must satisfy $0 < p < 1$; choosing $p = 0$ or $p = 1$ makes two of the four rates unestimable from first events.

6 Pure-Pool Splitting: a Multi-Cell First-Event Scheme

6.1 Motivation

In the single-cell first-event scheme (Section 5.6) each child pool starts from exactly one founder cell and contributes a single binary observation — effectively a Binomial(1, p) trial — to the estimation of $p = a/(a + b)$ (for X-seeded pools) or $q = c/(c + d)$ (for Y-seeded

pools). The information per pool is therefore low: regardless of how many cells the pool eventually contains, the observation reduces to a single coin flip about p.

The *pure-pool* scheme breaks this constraint by running each pure pool until the first cross-type event rather than stopping after one trial. The number of same-type events observed before that termination is itself informative about p, converting the per-pool data structure from a Binomial$(1, p)$ experiment to a NegativeBinomial$(1,\, 1-p)$ (equivalently, Geometric$(1-p)$) experiment.

Concretely, this is achieved by replacing single-cell founders with entire pure populations. When any pool first satisfies $\min(n_X,\, n_Y) \geq K$, it is dissolved and replaced by two *pure* pools:

- a **pure-X pool** containing all n_X cells of type X (and 0 cells of type Y), and
- a **pure-Y pool** containing all n_Y cells of type Y.

Each pure pool then evolves independently. Because a pure-X pool has no Y cells, every division event is unambiguously type a ($X \to X{+}X$) or type b ($X \to X{+}Y$): the event type is fully determined by the count change alone, without requiring individual event labels. The *pure phase* lasts until the first cross-type event ($X \to X{+}Y$ for a pure-X pool), after which the pool returns to mixed status and may eventually trigger a new split.

6.2 Information gain per pool

In a pure-X pool, successive events are independent Bernoulli trials with $P(\text{type } a) = a/(a+b)$, regardless of the current pool size (the pool size cancels from the competing rates $a\,n_X$ and $b\,n_X$). The pure phase ends at the first type-b event, so the number of type-a events $k_a \sim \text{Geometric}(b/(a+b))$ with mean $(a+b)/b$. For our parameters,

$$E[\text{events per pure-X pool}] = \frac{a+b}{b} = \frac{0.6}{0.1} = 6, \qquad E[\text{events per pure-Y pool}] = \frac{c+d}{c} = \frac{0.45}{0.05} = 9.$$

The pure-pool MLE benefits from *two* distinct information sources that are absent in the single-cell scheme.

1. **Count information (Negative Binomial).** The ratio $k_a/(k_a+1)$ is a consistent estimate of $p = a/(a+b)$ derived entirely from event counts, with no need for timing. This is the sufficient statistic of the NegBin$(1,\, b/(a+b))$ model. In the single-cell scheme $k_a \in \{0, 1\}$ and this ratio equals the single binary outcome — no richer than one coin flip.
2. **Timing information (Poisson process).** The integrated exposure Λ_X and the total event count $k_a + 1$ jointly estimate the overall rate $a + b$, exactly as in the standard CTMC MLE.

The combined MLE $\hat{a} = n_a/\Lambda_X$ uses both sources simultaneously. The Fisher information about $p_X = a/(a+b)$ per pool is therefore $\approx (a+b)/b = 6$ times larger than in the single-cell scheme (which always contributes exactly one Bernoulli trial).

6.3 Estimator

The likelihood in the pure phase factors as in Section 5. For all pure-X pools combined, the sufficient statistics are the total type-a event count n_a^{PP}, the total type-b event count n_b^{PP} (one per pool that completes its pure phase), and the total X-cell exposure $\Lambda_X^{\mathrm{PP}} = \sum_p \int n_X^{(p)}(t)\,dt$ over all pure-X phases. The MLE is identical in form to (13):

$$\hat{a} = \frac{n_a^{\mathrm{PP}}}{\Lambda_X^{\mathrm{PP}}}, \quad \hat{b} = \frac{n_b^{\mathrm{PP}}}{\Lambda_X^{\mathrm{PP}}}, \quad \hat{c} = \frac{n_c^{\mathrm{PP}}}{\Lambda_Y^{\mathrm{PP}}}, \quad \hat{d} = \frac{n_d^{\mathrm{PP}}}{\Lambda_Y^{\mathrm{PP}}}. \tag{18}$$

6.4 Simulation study and cost comparison

Table 7 and Figure 9 compare the pure-pool MLE against the single-cell first-event MLE and the full-data MLE for $K = 10$ and $N = 1000$ over $M = 200$ replicates.

Table 7: MLE comparison at $N = 1000$, $M = 200$ replicates. "FE" = single-cell first-event ($p = 0.5$); "PP" = pure-pool. The mean number of pure pools generated is 30 (FE: 355 single-cell pools) for $K = 10$, and 7 (FE: 170) for $K = 50$.

			First-event (K)		Pure-pool (K)	
Param	True	Full-data SD	Mean	SD	Mean	SD
$K = 10$						
a	0.500	0.020	0.511	0.053	0.516	0.063
b	0.100	0.008	0.103	0.022	0.108	0.027
c	0.050	0.010	0.050	0.020	0.057	0.020
d	0.400	0.027	0.401	0.057	0.428	0.052
$K = 50$						
a	0.500	0.020	0.505	0.069	0.520	0.141
b	0.100	0.008	0.102	0.030	0.142	0.116
c	0.050	0.010	0.050	0.025	0.080	0.081
d	0.400	0.027	0.410	0.075	0.411	0.108

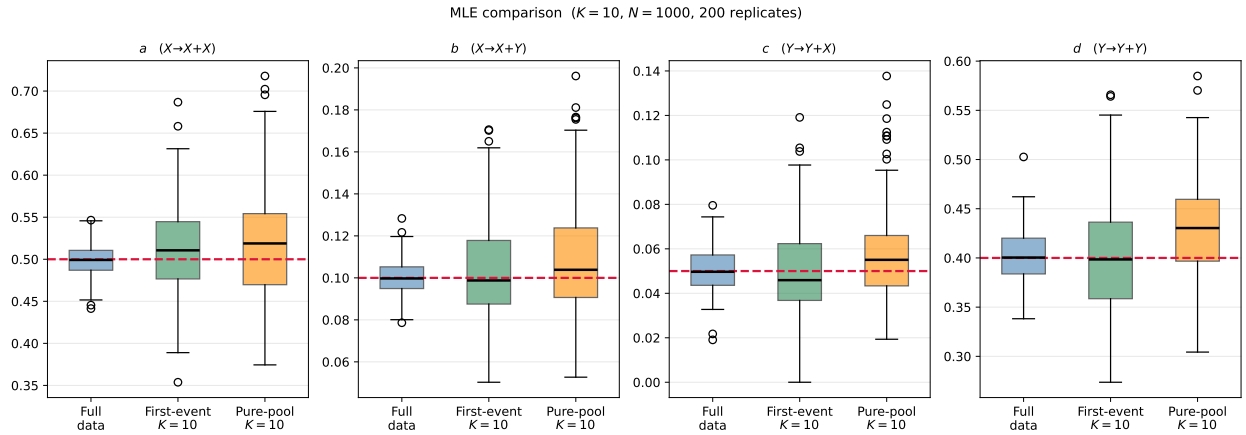

Figure 9: MLE comparison at $K = 10$, $N = 1000$, $M = 200$ replicates. Blue: full-data MLE (benchmark). Green: single-cell first-event MLE ($K = 10$, $p = 0.5$, 355 single-cell pools). Orange: pure-pool MLE ($K = 10$, 30 pure pools). Red dashed lines: true values.

6.5 Conclusions

- **At fixed N, single-cell first-event outperforms pure-pool.** Table 7 shows that for $K = 10$ and $N = 1000$, the single-cell first-event MLE has lower SD than the pure-pool MLE for three of four parameters (a, b, d). The reason is simple: the $\min(X, Y) \geq K$ trigger is harder to satisfy than total cells $= K$, so it fires far less often — generating only ≈ 30 pure pools vs. ≈ 355 single-cell pools. Fewer pools dominate even though each pure pool is individually more informative.
- **Each pure pool carries more information per split event.** In a pure-X pool every event is an unambiguous Bernoulli$(a/(a+b))$ trial; the pure phase lasts $(a+b)/b \approx 6$ events on average. This is $\approx 6\times$ the information of a single-cell pool (one trial). If the *number of split events* (e.g. experimental passages) is the binding cost rather than total cell count, the pure-pool scheme is preferable: it extracts more information from each split.
- **Large K is inadvisable.** At $K = 50$ only ≈ 7 pure pools are generated at $N = 1000$, giving substantially higher variance and noticeable finite-sample bias (see the $K = 50$ block of Table 7). Small K (e.g. $K = 5$–10) is recommended for the pure-pool scheme.

7 Trajectory-Based Estimation from Count Data

7.1 Motivation

Both the first-event and pure-pool schemes require tracking *individual events*: knowing whether a given cell division produced an X or Y offspring. In many experimental settings only aggregate cell counts are observable — for example, colony sizes counted under a microscope or by flow cytometry — with no ability to tag which cell generated which offspring.

We ask: can the four rates (a, b, c, d) be estimated accurately with far fewer pools by using the entire count trajectory $(n_X(t),\, n_Y(t))$ at a sequence of checkpoints, rather than just the first event?

7.2 The WLS estimator

The mean dynamics of the two-type branching process satisfy the linear ODE

$$\frac{d}{dt}\begin{pmatrix} E[X] \\ E[Y] \end{pmatrix} = \underbrace{\begin{pmatrix} a & c \\ b & d \end{pmatrix}}_{M} \begin{pmatrix} E[X] \\ E[Y] \end{pmatrix}. \tag{19}$$

Discretising over a short interval $[t_i,\, t_{i+1}]$ with midpoint counts $\bar{n}_X = (n_X(t_i) + n_X(t_{i+1}))/2$ and $\bar{n}_Y$ analogously gives the approximate linear model

$$\frac{\Delta X_i}{\Delta t_i} \approx a\,\bar{n}_{X,i} + c\,\bar{n}_{Y,i}, \tag{20}$$

$$\frac{\Delta Y_i}{\Delta t_i} \approx b\,\bar{n}_{X,i} + d\,\bar{n}_{Y,i}. \tag{21}$$

Because ΔX_i and ΔY_i are approximately Poisson with means $(a\bar{n}_X + c\bar{n}_Y)\Delta t_i$ and $(b\bar{n}_X + d\bar{n}_Y)\Delta t_i$, the variance of the observed rate $\Delta X_i/\Delta t_i$ scales as $1/\Delta t_i$. We therefore fit (20)–(21) by *weighted* least squares (WLS) with weights Δt_i, implemented as non-negative least squares (NNLS) to enforce $a, b, c, d \geq 0$.

Given P independent pools each sampled at M equally-spaced (by total cell count) checkpoints, the two $P(M-1)$-row linear systems are solved simultaneously, yielding estimates $(\hat{a}, \hat{b}, \hat{c}, \hat{d})$. No individual event labels are required — only the two cell counts at each checkpoint.

7.3 Identifiability

Equation (20) identifies a and c separately only when $\bar{n}_{Y,i} > 0$ for some intervals, i.e. only after Y cells have appeared. Starting from $(n_X, n_Y) = (1, 0)$, Y cells first appear (in expectation) after $1/b = 10$ X events. With $M = 50$ checkpoints over a $N = 1000$-cell simulation the pool has typically reached several hundred cells of each type by the later checkpoints, providing ample variation in $(\bar{n}_X, \bar{n}_Y)$. A single pool ($P = 1$) is therefore *not* sufficient: the c (Y→X) and d (Y→Y) rates, which are driven by rare early-Y intervals, are estimated with large variance and bias. Multiple pools ($P \geq 3$) supply independent trajectory realisations and regularise the design matrix.

7.4 Simulation study

Table 8 and Figure 10 compare the trajectory WLS estimator against the full-data MLE and the single-cell first-event MLE ($K = 10$, ≈ 355 pools) over $M_{\text{rep}} = 200$ replicates, $N = 1000$. Three values of P are evaluated; each pool provides $M = 50$ checkpoints.

Table 8: Trajectory WLS versus first-event MLE ($N = 1000$, $M_{\text{rep}} = 200$). Traj_P = trajectory WLS using P pools with 50 snapshots each. Full-data SD is the benchmark (uses all event labels).

			First-event ($K = 10$)		Traj_1		Traj_3		Traj_{10}	
Param	True	FD SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
a	0.500	0.020	0.511	0.053	0.402	0.148	0.489	0.054	0.503	0.022
b	0.100	0.008	0.103	0.022	0.120	0.123	0.091	0.050	0.091	0.020
c	0.050	0.010	0.050	0.020	0.215	0.248	0.066	0.085	0.043	0.033
d	0.400	0.027	0.401	0.057	0.386	0.195	0.417	0.083	0.414	0.035

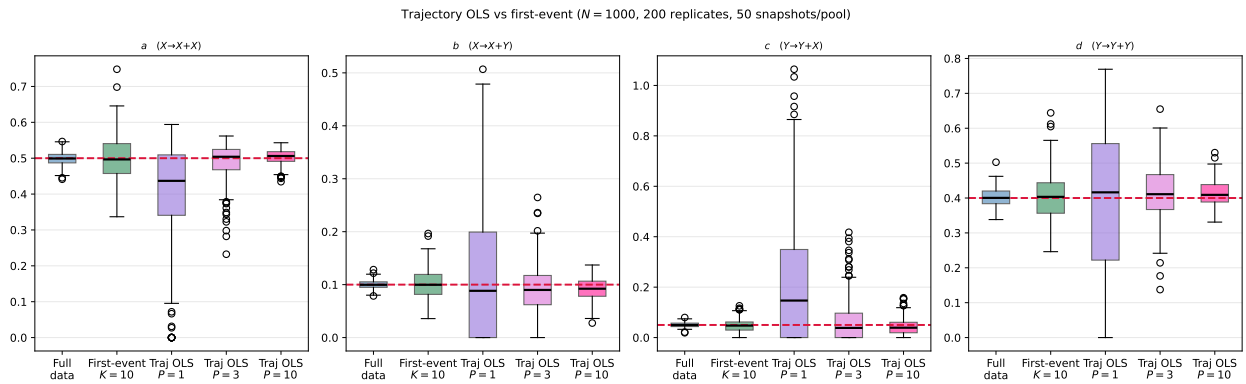

Figure 10: Trajectory WLS versus first-event MLE ($N = 1000$, $M_{\text{rep}} = 200$, 50 snapshots per pool). Blue: full-data MLE. Green: first-event ($K = 10$, ≈ 355 pools). Purple/orchid/pink: trajectory WLS with $P = 1$, 3, 10 pools. Red dashed lines: true values.

7.5 Conclusions

- $P = 1$ **pool is insufficient.** A single trajectory starting from $(1, 0)$ is nearly unidentifiable for c (Y→X): the design matrix for (20) is nearly rank-deficient until Y cells are abundant, leading to large variance and upward bias ($\hat{c} \approx 0.22$ vs. true 0.05).

- $P = 3$ **pools achieves accuracy comparable to first-event for** a **and** d**.** SD for a is 0.054 (FE: 0.053); SD for d is 0.083 (FE: 0.057). The rarer rates b and c remain noisier because they depend on the early-appearance of minority cells. Three pools reduce pool count from 355 to 3 — a 120× reduction.

- $P = 10$ **pools matches or beats first-event for all parameters except** c**.** SD for a is 0.022 (FE: 0.053, full-data: 0.020); SD for b is 0.020 (FE: 0.022); SD for d is 0.035 (FE: 0.057). Only c remains slightly worse (SD 0.033 vs. FE 0.020), because c governs the rare Y→X events which are underrepresented when starting from a single X cell. Ten pools achieve a 35× reduction in pool count relative to first-event.

- **Practical recommendation (at** $M = 50$**).** If experimental cost is proportional to pool count and individual event labels are unavailable, trajectory WLS with $P = 10$

pools and $M = 50$ count snapshots per pool is preferred. The estimator is more complex than first-event (requires solving two linear systems rather than computing a ratio), but requires only a single count observation at each checkpoint — no cell tracking or sorting needed.

7.6 Effect of checkpoint density M

A key practical question is how many checkpoints per pool are needed. Fewer checkpoints reduce measurement burden but increase the step size Δt_i, which makes the mean-ODE approximation less accurate. Table 9 and Figure 11 show results for $P = 10$ pools and $M \in \{5, 10, 20, 50\}$ over 200 replicates.

Table 9: Trajectory WLS: effect of checkpoint density M at fixed $P = 10$ pools ($N = 1000$, 200 replicates). FE SD = first-event ($K = 10$, ≈ 355 pools) standard deviation.

			$M = 5$		$M = 10$		$M = 20$		$M = 50$	
Param	True	FE SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
a	0.500	0.053	0.421	0.060	0.474	0.036	0.494	0.027	0.502	0.024
b	0.100	0.022	0.063	0.029	0.080	0.022	0.088	0.020	0.091	0.020
c	0.050	0.020	0.117	0.106	0.079	0.061	0.056	0.043	0.048	0.037
d	0.400	0.057	0.419	0.053	0.424	0.040	0.418	0.035	0.414	0.034

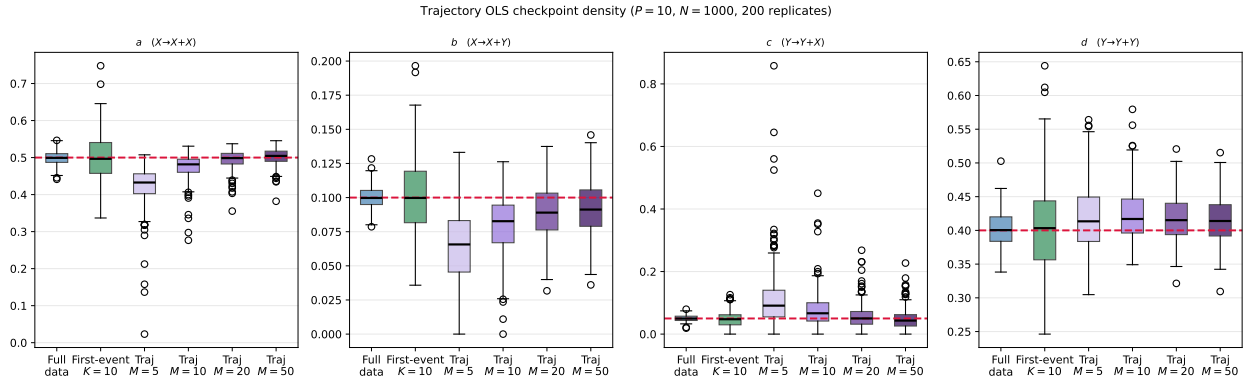

Figure 11: Trajectory WLS checkpoint-density sweep ($P = 10$, $N = 1000$, 200 replicates). Blue: full-data MLE. Green: first-event ($K = 10$). Purple shades (light to dark): trajectory WLS with $M = 5, 10, 20, 50$ checkpoints per pool. Red dashed: true values.

The dominant pattern is that bias — not variance — limits accuracy at small M. Table 10 decomposes the error for all four rates into bias and SD.

Table 10: Bias and SD of trajectory WLS for all rates, $P = 10$, $N = 1000$, 200 replicates. Bias $= \hat{\theta} - \theta_{\text{true}}$.

	a (true 0.500)		b (true 0.100)		c (true 0.050)		d (true 0.400)	
M	Bias	SD	Bias	SD	Bias	SD	Bias	SD
5	−0.079	0.060	−0.037	0.029	+0.067	0.106	+0.019	0.053
10	−0.026	0.036	−0.020	0.022	+0.029	0.061	+0.024	0.040
20	−0.006	0.027	−0.012	0.020	+0.006	0.043	+0.018	0.035
50	+0.002	0.024	−0.009	0.020	−0.002	0.037	+0.014	0.034

The bias arises because the mean-ODE linearisation $\Delta X \approx (a\bar{n}_X + c\bar{n}_Y)\,\Delta t$ is only exact for infinitesimal Δt; at coarse resolution the growing pool size within each interval introduces a systematic approximation error. The direction of the bias is consistent: a and b are underestimated (fewer X→X and X→Y events per unit exposure than predicted by the midpoint count) while c is overestimated.

Three findings stand out.

- $M = 5$ **is too coarse.** Bias dominates for all rates. The estimator carries more systematic error than a first-event MLE using 355 labelled pools.
- $M = 10$ **is a workable minimum when bias is acceptable.** SD falls near or below first-event for a (0.036 vs. 0.053) and d (0.040 vs. 0.057), but the remaining bias (-0.026 for a, $+0.029$ for c) is systematic and does not shrink with more replicates.
- $M = 20$ **is the practical sweet spot.** Bias falls to $\leq 1\%$ of the true value for all four rates, and SD is comparable to first-event. Moving to $M = 50$ yields only marginal further improvement. The scheme requires 10 pools with 20 count snapshots each — versus 355 event-labelled pools — a 35× reduction in pool count with no loss of accuracy.

8 Discussion

8.1 Founder effect and bottlenecks

The dominant feature of the splitting simulation is a manifestation of the *founder effect*: when a new pool is established from a single individual, the type of the founder disproportionately shapes the descendant population. With $K = 100$ this effect is weak, because each pool grows long enough for the composition to approach the eigenvector equilibrium before the next split. In the parallel setting, coexistence of many pools at every instant further averages out individual founder biases.

8.2 Invariance of growth rate

Because splits only redistribute existing cells (no net creation or destruction), the dominant eigenvalue λ_1 and therefore the exponential growth rate of the total population are the same in all scenarios. What changes is how quickly the composition vector $(X, Y)/N$ approaches the Perron eigenvector $\approx (0.578, 0.422)$.

8.3 Practical interpretation

If this model represents, for example, stem-cell passage in a culture dish, the splitting threshold K is the passage size and p is the fraction of stem cells (type X) deliberately maintained during passaging. The analysis shows that:

- Passaging at a large K is more "self-correcting" — the culture naturally drifts toward equilibrium composition.
- Parallel passaging (growing multiple sub-cultures simultaneously) greatly suppresses composition drift and renders the result nearly independent of the seeding protocol.
- Rate parameters a, b, c, d can be estimated from the cell-count trajectory by the closed-form MLEs (13), without requiring any additional experimental interventions. The estimators are practically unbiased and have comparable precision under both protocols for $N = 1000$ cells observed.

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